

## Summary for: Edge AI applications in Traditional Businesses

### Introduction:

Edge AI, a combination of edge computing and artificial intelligence, involves processing data generated by Internet of Things (IoT) devices close to the source rather than relying on cloud-based systems. This technology is gaining traction in traditional businesses due to its ability to provide real-time insights, reduce latency, and enhance data privacy [1].

### Body:

#### Applications of Edge AI in Traditional Businesses:

1. Manufacturing: Edge AI is being used to optimize production processes, detect anomalies, and perform predictive maintenance. For instance, Siemens uses Edge AI to monitor its production lines in real time, allowing for immediate action in case of any irregularities [2].
2. Retail: Edge AI is revolutionizing the retail industry by enhancing customer experiences and streamlining operations. For example, Amazon Go uses Edge AI for its cashier-less stores, where customers can walk in, pick up items, and walk out without having to wait in line [3].
3. Healthcare: Edge AI is improving patient care by enabling real-time monitoring and diagnostics. Philips, for instance, uses Edge AI in its eICU program to provide continuous

patient monitoring, thereby improving outcomes [4].

4. Agriculture: Traditional farming is being transformed with the help of Edge AI. Companies like John Deere are using this technology for precision farming, which includes soil analysis, crop health monitoring, and predictive analytics for weather conditions [5].

#### Conclusion:

Edge AI holds significant promise for traditional businesses across various sectors, from manufacturing and retail to healthcare and agriculture. By leveraging this technology, businesses can not only enhance operational efficiency but also improve customer experience and outcomes. However, the adoption of Edge AI also brings challenges, including data security and infrastructure requirements. Therefore, businesses need to carefully consider these factors before integrating Edge AI into their operations.

#### Sources:

[1] "What is Edge AI? Why Does It Matter?"

<https://www.forbes.com/sites/forbestechcouncil/2020/07/29/what-is-edge-ai-why-does-it-matter/?sh=6e3d687e1e12>

[2] "Siemens leverages Edge AI for Industrial Applications"

<https://www.siemens.com/global/en/home/products/automation/topic-areas/industrial-edge.html>

[3] "How Amazon Go Uses Edge AI"

<https://www.zdnet.com/article/how-amazon-go-uses-edge-ai/>

[4] "Philips eICU Program Leverages Edge AI"

<https://www.usa.philips.com/healthcare/consulting/articles/customer-story/eicu-program>

[5] "John Deere's Use of AI in Agriculture"

<https://www.deere.com/en/technology-products/precision-ag-technology/>

#### Analysis:

##### Key Actionable Insights:

- Manufacturing, retail, healthcare, and agriculture sectors are leveraging Edge AI for a variety of applications, which include real-time monitoring, anomaly detection, predictive maintenance, customer experience enhancement, and precision farming.
- Businesses are integrating Edge AI to improve operational efficiency, enhance customer experience, and achieve better outcomes.
- Despite its benefits, the adoption of Edge AI presents challenges related to data security and infrastructure requirements.

##### Suggested Next Steps or Projects:

- Conduct a feasibility study to assess the suitability of Edge AI in your business operations considering both the benefits and challenges.
- Develop a pilot project to measure the impact of Edge AI on a specific aspect of your business, such as operational efficiency or customer experience.
- Plan training programs for employees to ensure they have the necessary skills to work with Edge AI systems.
- Explore partnerships with AI technology providers to leverage their expertise and experience in implementing Edge AI solutions.

### Related Areas/Topics Worth Exploring Further:

- Investigate the ethical implications of using Edge AI, focusing on aspects such as data privacy and algorithmic bias.
- Research how Edge AI is being used in other industries beyond the ones mentioned, such as logistics, finance, or construction, to gain a broader understanding of its applications.
- Study the latest advancements in Edge AI technology to stay updated with its potential future applications.
- Examine the regulatory landscape related to the use of Edge AI in business, as this could have significant implications for its adoption.